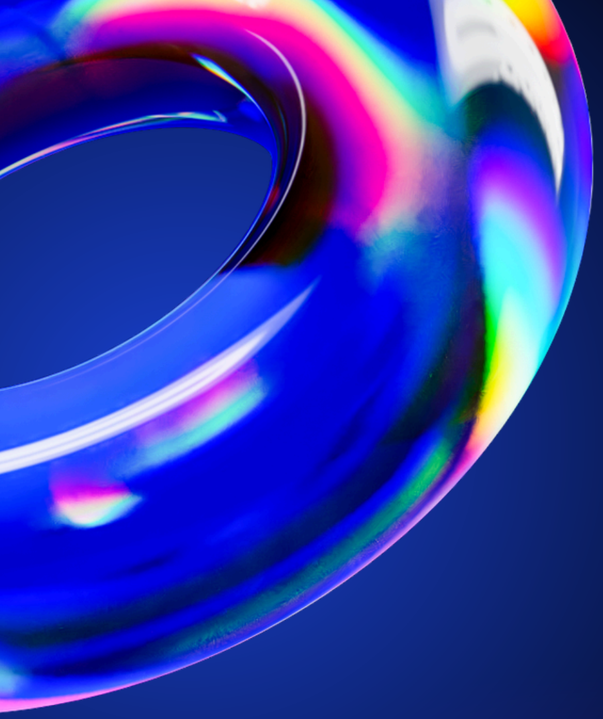
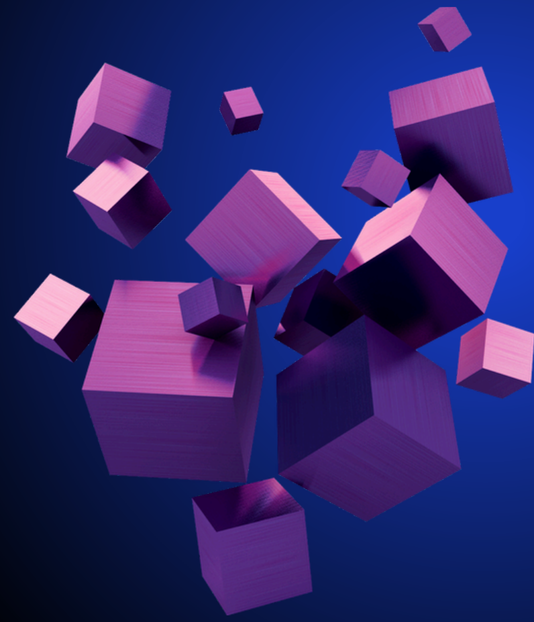


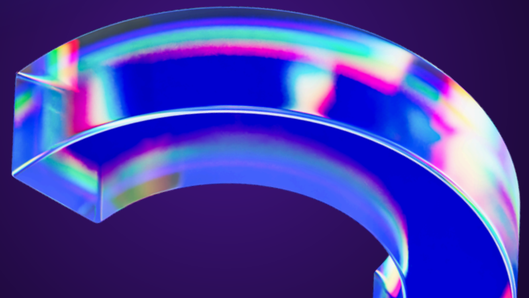
Holo Bench



Immersive Lego-Simulation Bench

Maymoona Khan





Me

- Junior in CECS
- Hometown: LA
- 3rd year in Makers!
- **Hobbies**: Legos 👀, baking, pokemon, rollerblading, movies, climbing (in progress..)







Me

❤️ to hang w/ friends!



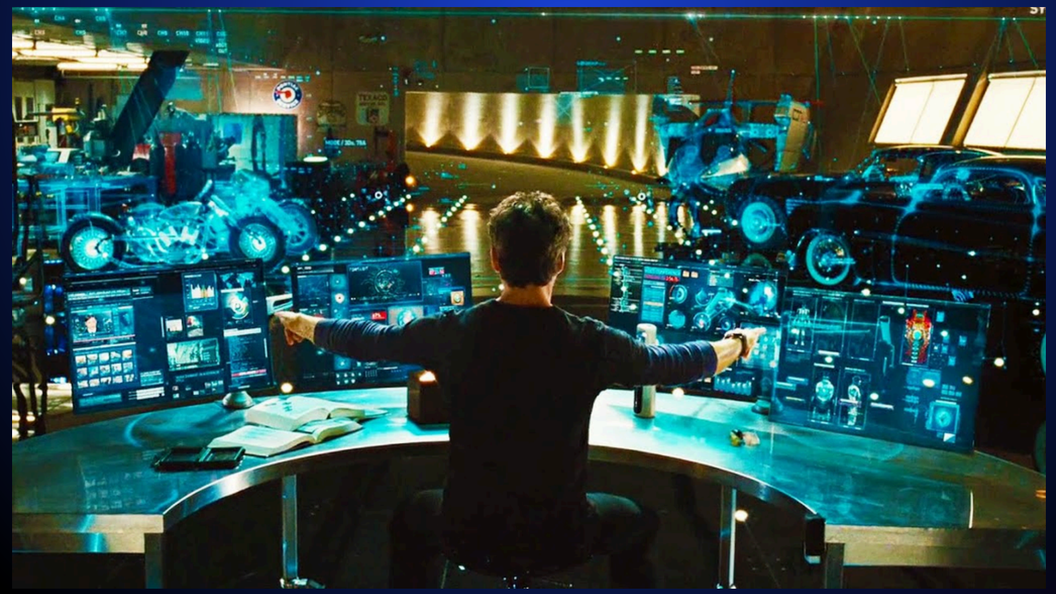


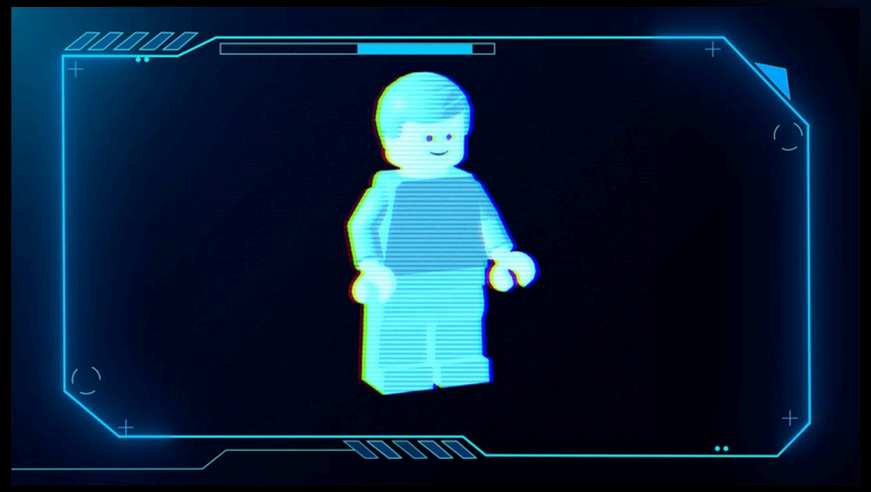




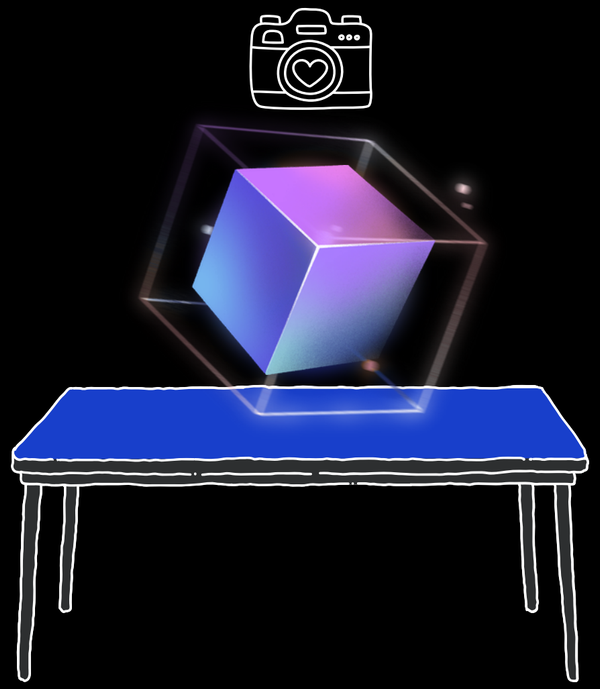
Inspiration







Introducing....







3-D Holographic Lego-Building Bench!

- Projector display that renders 3-D images of lego pieces that the user can select, drag, and snap together.
- Gesture and hand-controled with CV and EMG gloves

Subteams

Electrical
Gloves Team

- Design sensor circuit for gloves (EMG+flex sensors)
- Wiring, PCB work, signal and sensor work

CS
Computer Vision & App Team

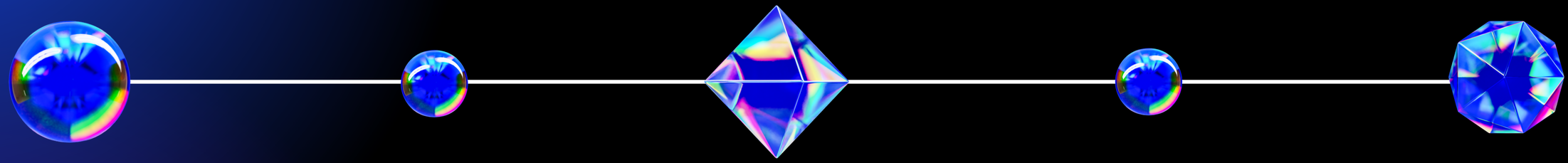
- Builds the CV hand-gesture tracking
- 3D image rendering
- Orchestration logic to fuse CV position + Glove grab/release

Mechanical
Display System Team

- Build the rear-projection display rig (mount, film, projector, light-control enclosure)
- Mounting hardware for the camera



Timeline



Start

Mid-Sem

- CV team: can track a marker's position live
- Glove team: detect a "did a contraction happen" event with basic sensors

Fall Showcase

- Simple Lego pieces can be dragged/rotated on screen

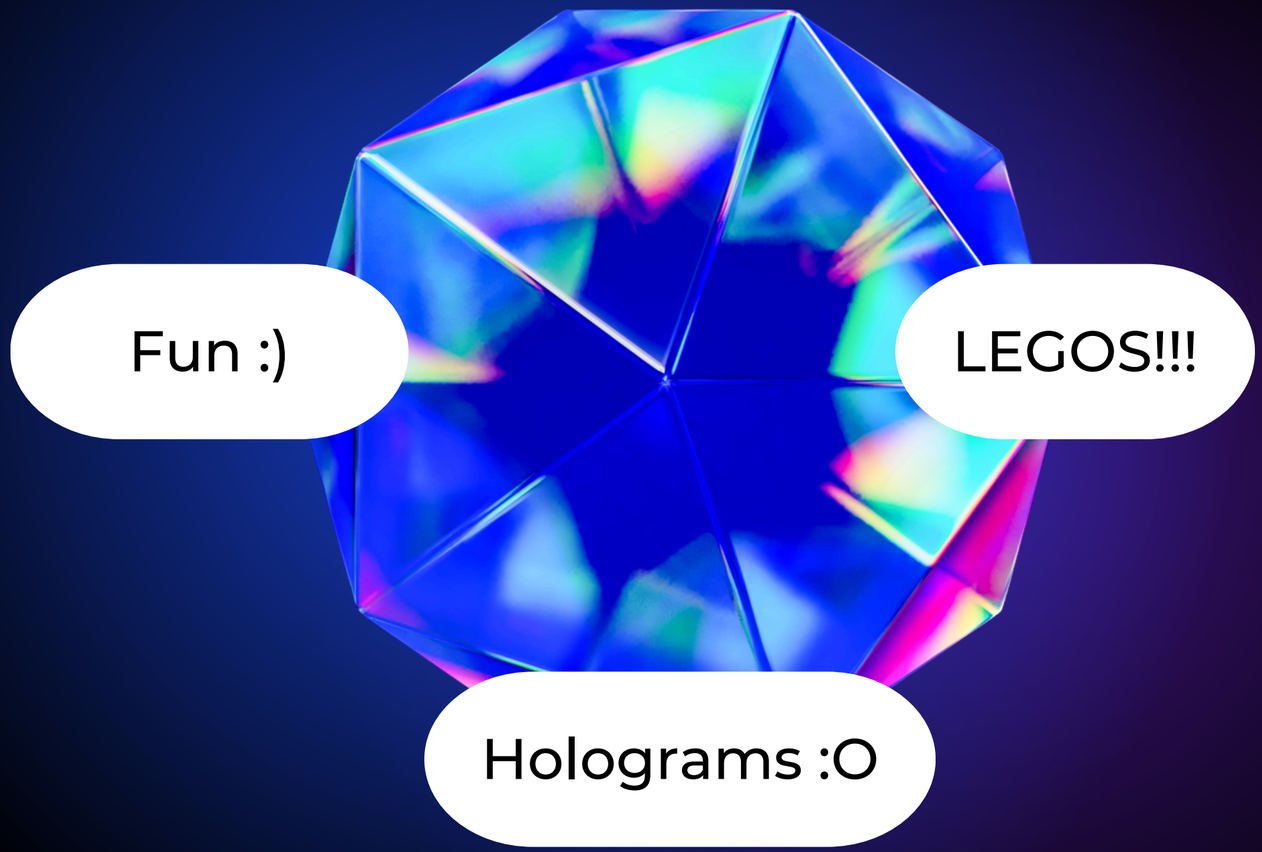
Mid-Sem

- CV + glove work together at a basic level

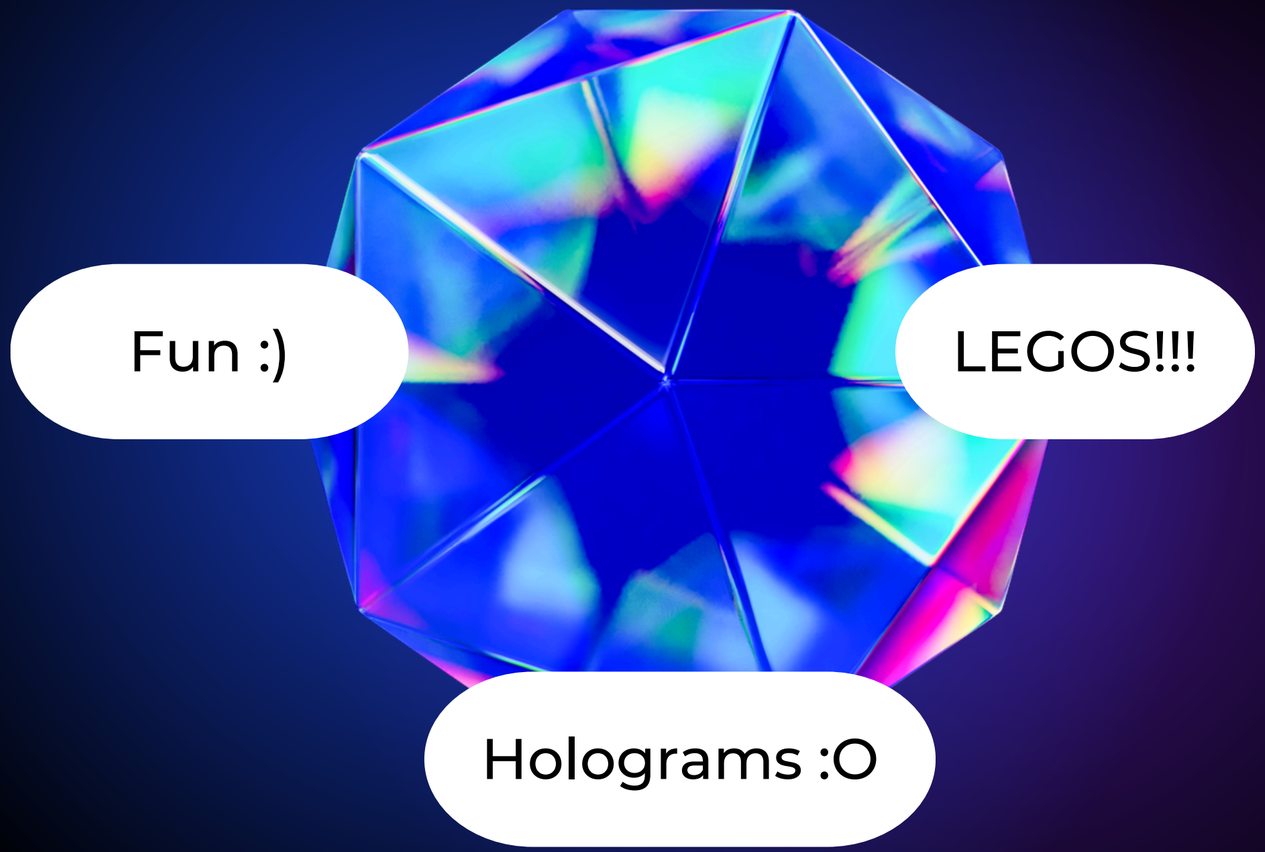
Spring Showcase

- CV + glove work together to select, drag, drop, and snap a full Lego set into place.

Why you should join!



Why you should join!



- working with a Qualcomm Board
- YOU get to design and pitch in ideas as well!
- opportunity to build and learn lots of skills!

Why you should join!

You ->



Questions?
Ideas?
Suggestions?

Feel Free to Reach out!!

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818-941-2517